

JAVA PROGRAMMING

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Course Code : CSA0921

1.

Q1 Handling ArithmeticException

10 marks

Write a Java program to divide two integers entered by the user. If the denominator is zero, handle the exception using a try-catch block and display an appropriate error message.

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         try {
6             int a = sc.nextInt();
7             int b = sc.nextInt();
8
9             System.out.println("Result = " + (a / b));
10        } catch (ArithmeticException e) {
11            System.out.println("Error: Cannot divide by zero.");
12        }
13        sc.close();
14    }
15 }
```

Input

20
5

Output

Result = 4

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2.

Q2 Handling ArrayIndexOutOfBoundsException

10 marks

Write a Java program to create an integer array of five elements. Ask the user to enter an index and display the corresponding element. If the index is outside the valid range, handle the exception and display an error message.

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int[] arr = {10, 20, 30, 40, 50};
6         try {
7             System.out.print("Enter index (0 to 4): ");
8             int index = sc.nextInt();
9             System.out.println("Element = " + arr[index]);
10        }
11        catch (ArrayIndexOutOfBoundsException e) {
12            System.out.println("Error: Invalid array index.");
13        }
14        sc.close();
15    }
16 }
```

Input

7

Output

```
Enter index (0 to 4): Error:
Invalid array index.
```

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3.

Q3 Handling NumberFormatException

10 marks

Write a Java program to accept a number as a string and convert it into an integer using Integer.parseInt(). If the input is not a valid integer, handle the exception and display an appropriate message.

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         try {
6             System.out.print("Enter a number: ");
7             String str = sc.nextLine();
8             int num = Integer.parseInt(str);
9             System.out.println("Integer value = " + num);
10        }
11        catch (NumberFormatException e) {
12            System.out.println("Error: Invalid integer.");
13        }
14        sc.close();
15    }
16 }
```

Input

123

Output

```
Enter a number: Integer value = 123
```

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4.

Your Code

```
1 import java.util.Scanner;
2 class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int[] arr = new int[3];
6         try {
7             System.out.print("Enter first number: ");
8             int a = sc.nextInt();
9             System.out.print("Enter second number: ");
10            int b = sc.nextInt();
11            int result = a / b;
12            System.out.print("Enter array index (0 to 2): ");
13            int index = sc.nextInt();
14            arr[index] = result;
15            System.out.println("Result stored in array = " + arr[index]);
16        } catch (ArithmeticException e) {
17            System.out.println("Error: Cannot divide by zero.");
18        } catch (ArrayIndexOutOfBoundsException e) {
19            System.out.println("Error: Invalid array index.");
20        }
21        sc.close();
22    }
23 }
```

Input

20
5
1

Output

Enter first number: Enter second
number: Enter array index (0 to 2):
Result stored in array = 4

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